

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

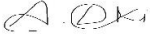
Annexure C

Case study

<i>Criteria</i>	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						

Code of Conduct:

*I A. LOKESH KUMAR(192524157) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP,FP,FN,TN and Calculate the Precision and Recall and F1 Score .

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains "Offer" (Yes/No)
- Contains "Free" (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

1.

A Shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

Sol.)

TD, FP, FN, TN

		pred	
		W = yes	W = No
Act	W = y (P)	TP Wolf came Warning	FN Wolf came No warning
	W = N (N)	FP Wolf No come but warning	TN No Wolf came No warning

Prediction.

Actual

2.

Assume there are 100 images, 30 of them depict a cat the rest do not. A machine learning model predicts the occurrences of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images

Sol:

		Pred	
		Cat = Positive	No cat = Positive
Act	Cat = Positive	TP = 25 (Actual Cat & Predicted Cat)	FN = 25 (Actual cat but predicted no cat)
	No cat (Negative)	FP = 20 (Act No cat but Predicted cat)	TN = 50 (Act No cat & Predicted No cat)

Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$= \frac{25}{25 + 20} = 0.556 \times 100 = 55.6\%$$

Recall:

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP+FN} \\ &= \frac{85}{2545} \\ &= 0.8333 \times 100 \\ &= 83.33\%\end{aligned}$$

F1 Score:

$$\begin{aligned}\text{F1 Score} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= \frac{2 \times 0.556 \times 83.33}{0.556 + 83.33} \\ &= 66.67\%\end{aligned}$$

3

An E-mail Service Provider wants to classify emails as spam or Not spam using features

- Contains "offer" (Yes/No)
- Contains "free" (Yes/No)
- Email / Length (Short/Long).

Sol:

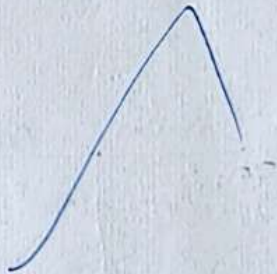
1. Total = 6

Spam = 3

Not spam = 3

$P(\text{Spam}) = \frac{3}{6} = \frac{1}{2}$

$P(\text{Not spam}) = \frac{3}{6} = \frac{1}{2}$



2. Conditional Probability Table.

$P(O = \text{Yes} / \text{spam}) = \frac{2}{3}$	$P(O = \text{Yes} / \text{No spam}) = \frac{1}{3}$
$P(O = \text{No} / \text{spam}) = \frac{1}{3}$	$P(O = \text{No} / \text{No spam}) = \frac{2}{3}$
$P(F = \text{Yes} / \text{spam}) = \frac{2}{3}$	$P(F = \text{Yes} / \text{No spam}) = \frac{1}{3}$
$P(F = \text{No} / \text{spam}) = \frac{1}{3}$	$P(F = \text{No} / \text{No spam}) = \frac{2}{3}$
$P(L = \text{Yes} / \text{spam}) = \frac{2}{3}$	$P(L = \text{Yes} / \text{No spam}) = \frac{1}{3}$
$P(L = \text{No} / \text{spam}) = \frac{1}{3}$	$P(L = \text{No} / \text{No spam}) = \frac{2}{3}$

3. Using Naive Bayes, Classify :- $P(A/B) = \frac{P(BA)P(A)}{P(B)}$

Offer	Free	Length	Class
Yes	No	short	Spam
Yes	No	Long	Not spam
No	Yes	short	Spam
No	Yes	long	Not spam
Yes	Yes	long	Spam
No	No	short	Not spam

For Spam: $P(\text{spam} \times \text{spam}) \propto P(\text{Yes/spam}) \times P(\text{No spam}) \times P(\text{short} / \text{spam})$

$$\frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} \times \frac{2}{3} = 0$$

For Not Spam: $P(\text{Not spam}) \times P(\text{Not spam}) \times P(\text{Yes/Not spam}) \times P(\text{No/Not spam}) \times P(\text{short / Not spam})$

$$\frac{1}{2} \times \frac{1}{3} \times \frac{1}{3} \times \frac{1}{2} = \frac{1}{18}$$

= 0.056 for Not spam

$$\therefore P(\text{Not spam } x) > P(\text{spam } x)$$

Therefore using the Naive Bayes classifies the given email (offer = yes, free = no, length short) is classified as Not spam because its probability for Not spam is higher than spam.

(iv) Comparing results using Naive Bayes

Dataset	(i) Bayes	(ii) Bayes
Fries	(100) 93.20	93.73
Super market	(100) 63.71	63.71
	(10/10)	(10/10)

Key (i) Bayes Naive Bayes
(ii) Bayes Naive Bayes



Actual class Predicted class	Cancer = Yes	Cancer = No	Total
Cancer = Yes	90	210	300
Cancer = No	110	9560	9700
Total	200	9770	10000

1. Accuracy = $\frac{TP + TN}{R}$

$$= \frac{90 + 9560}{10000} = 0.965 \times 100 = 96.5\%$$

2. Error Rate = $1 - \text{accuracy}$

$$= 1 - 96.5\%$$

$$= 3.5\%$$

3. Sensitivity = $\frac{TP}{P}$

$$= \frac{90}{200} \times 100$$

$$= 45\%$$

4. Specificity = $\frac{TN}{N}$

$$= \frac{9560}{9700}$$

$$= 0.985 \times 100$$

$$= 98.5\%$$

$$\text{Precision: } \frac{TP}{TP + FP}$$

$$= \frac{90}{90 + 40}$$

$$= \frac{90}{130}$$

$$= 0.6923$$

$$= 69.23\%$$

$$\text{Recall: } \frac{TP}{TP + FN}$$

$$= \frac{90}{90 + 60} = \frac{90}{150} \times 100 = 60$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 69.23 \times 60}{69.23 + 60} = \frac{8307.6}{129.23} = 64.28\%$$

$$F_1 = 64.28\%$$

Actual \ Prediction		
	Actual Cat	Actual No Cat
Predicted Cat	TP	FP
Predicted No Cat	FN	TN

2.)

Predicted Cat

TP = 25

FP = 20

Predicted No Cat

FN = 5

TN = 50

$$\begin{aligned}\text{Precision} &= \frac{TP}{TP+FP} \\ &= \frac{25}{25+20} \\ &= \frac{25}{45} = 0.556\end{aligned}$$

$$\text{Precision} = 55.56\%$$

Recall:

$$\begin{aligned}\text{Recall} &= \frac{TP}{TP+FN} \\ &= \frac{25}{25+5} = \frac{25}{30} = 0.833\end{aligned}$$

$$\boxed{\text{Recall} = 0.833}$$

F1 Score:

$$F1 = 2 \times$$

$$\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= 2 \times \frac{0.556 \times 0.833}{0.556 + 0.833}$$

$$= 0.667$$

$$F1 \text{ Score} = 66.67\%$$

3)

Conditional Probability

Total Emails: 6

Spam: 3

Not Spam: 3

$$P(\text{Spam}) = \frac{3}{6} = 0.5$$

$$P(\text{Not Spam}) = \frac{3}{6} = 0.5$$

$$P(\text{offer} = \text{Yes} / \text{spam}) = \frac{2}{5}$$

$$P(\text{free} = \text{Yes} / \text{spam}) = \frac{3}{5} = 1$$

$$P(\text{length} = \text{short} / \text{spam}) = \frac{2}{5}$$

For Not Spam:

$$P(\text{offer} = \text{Yes} / \text{Not spam}) = \frac{1}{3}$$

$$P(\text{free} = \text{Yes} / \text{Not spam}) = \frac{0}{3} = 0$$

$$P(\text{length} = \text{short} / \text{Not spam}) = \frac{1}{3}$$

Chasid
12/10/20